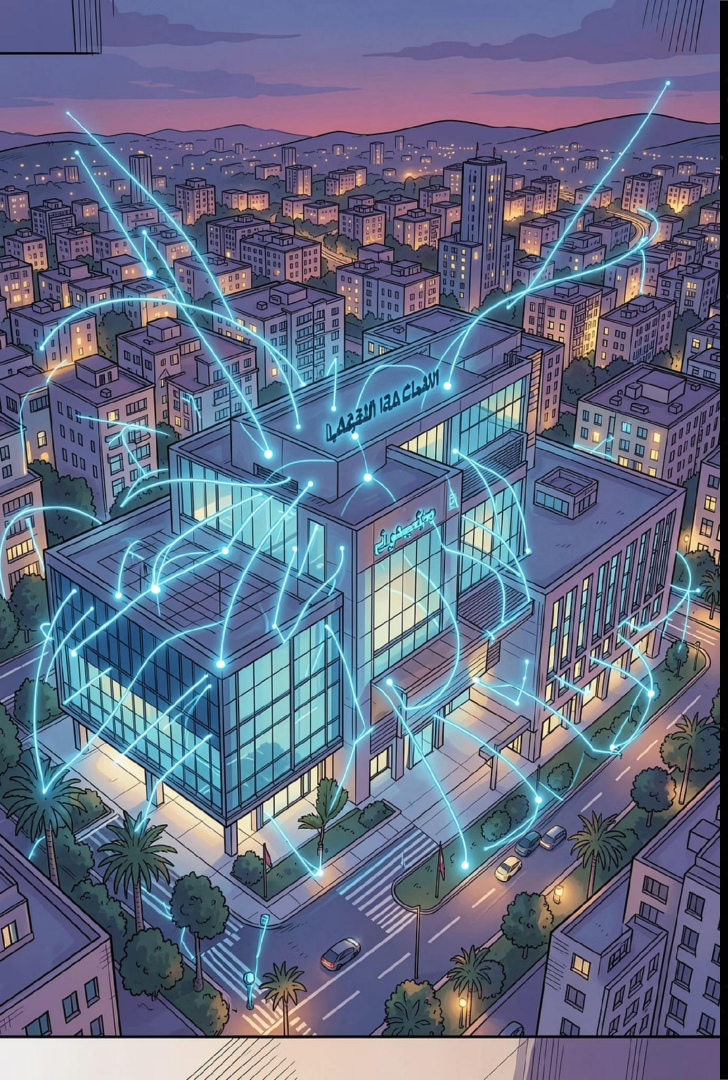


E-Government Platform for Lebanon

A university capstone project digitizing the submission, review, and payment stages of official document requests — reducing in-person attendance to the single legally necessary step of final collection.

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The Problem

Access to government services in Lebanon has long depended on **physical presence**. Obtaining a birth certificate, renewing a driving license, or registering a business requires traveling to a specific office, waiting in line, submitting paper documents, and returning later to collect the result.

Travel Cost

Citizens outside major cities bear disproportionate time and expense.

Lost Productivity

Multiple visits required for a single document request.

Office Overload

Routine administrative work concentrates in offices not equipped to handle it efficiently.

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Project Objectives

The central objective is to **reduce the need for in-person attendance** by providing a secure, remote channel for citizens to submit and track applications.

1

Single Access Point

One mobile application for requesting multiple categories of government documents.

2

Controlled Admin Review

A web dashboard for consistent, auditable review with decision attribution and timestamps.

3

Secure Identity & Payment

OTP-gated registration, memory-hard password hashing, and protected financial transactions.

4

Faithful Process Modeling

The digital process reflects the real administrative process — not a simplified version.

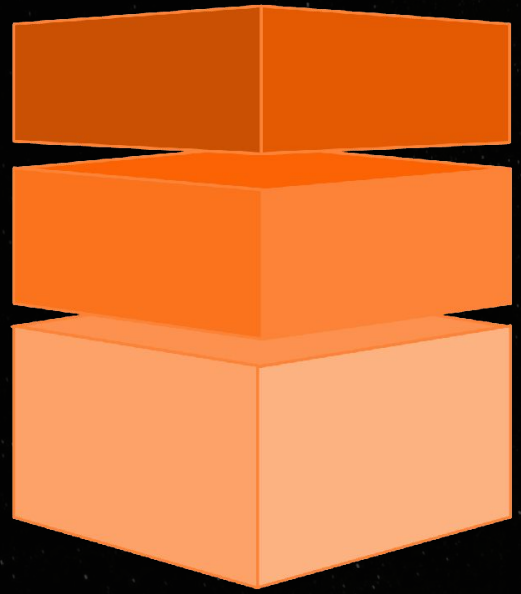
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System Architecture

A **three-tier architecture** with a RESTful API as the sole bridge between clients and data. Neither client communicates directly with the database.

Presentation Tier

Flutter app for citizens and Angular admin dashboard.



Application Tier

NestJS REST API with JWT auth, Twilio, Stripe integrations.

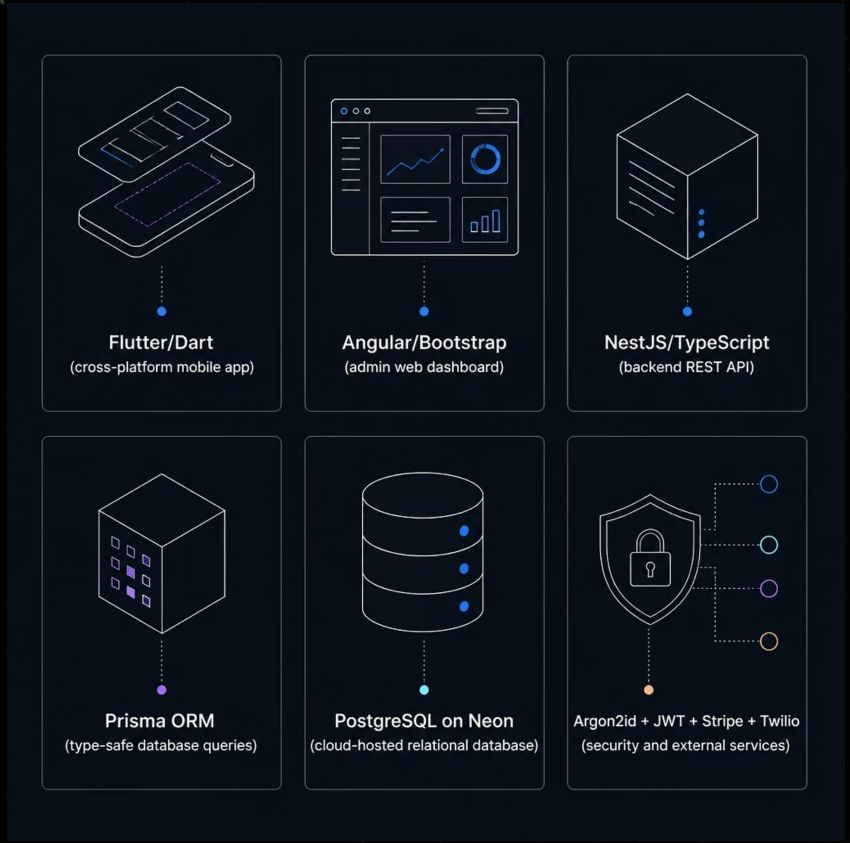
Data Tier

PostgreSQL hosted on Neon as the cloud database.

on authentication, the backend issues a signed JWT carrying the user's role. The database is a PostgreSQL instance hosted on Neon, ensuring cloud availability. Twilio Verify handles OTP delivery; Stripe processes payments — both integrations are strictly server-side.

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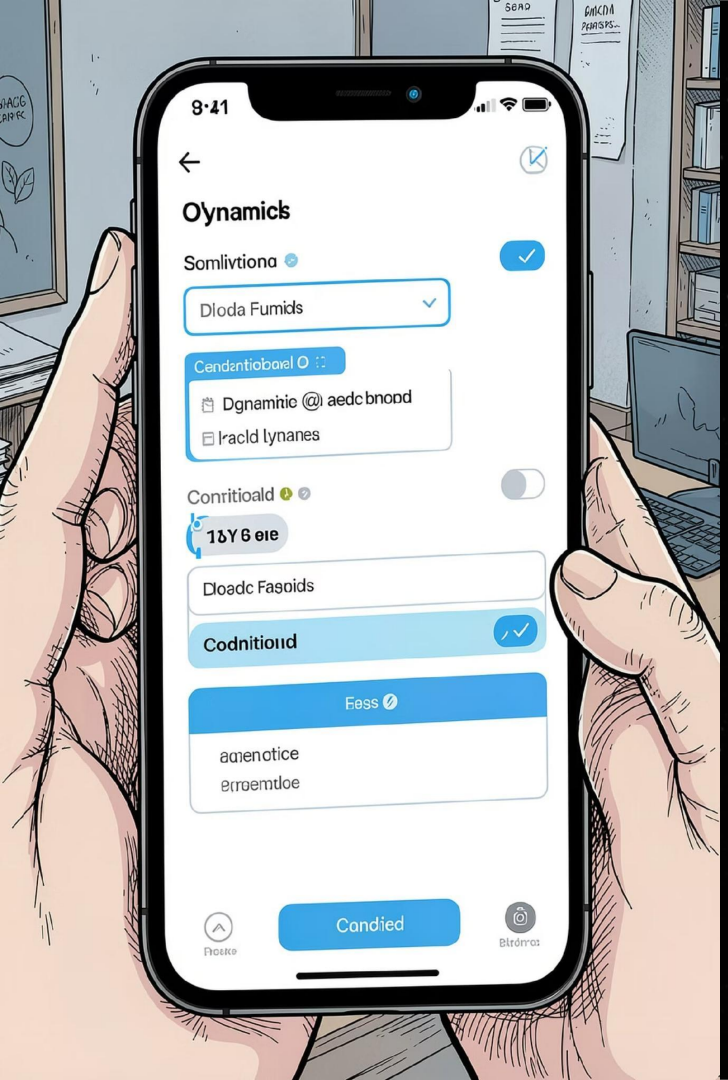
Technology Stack



Key Design Choices

- Argon2id — OWASP-recommended memory-hard password hashing
- JWT — Stateless, role-bearing authentication tokens
- Prisma — Schema introspected from live DB; constraints remain authoritative
- Neon — Serverless PostgreSQL, reachable from all environments
- Stripe — Server-side PaymentIntent, native PaymentSheet on mobile
- Twilio Verify — OTP delivery for registration, phone changes, and password resets

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Data-Driven Document System

Core innovation: Document requirements — input types, conditional reveal rules, file-upload flags — are **stored as data**, not hardcoded. New document types are introduced without code changes.

Requirements as Data

Each document's form is generated dynamically from admin-defined requirement records.

Conditional Fields

Fields appear only when triggering conditions are met — reducing citizen confusion.

Extensible Catalogue

Administrators add, edit, and deactivate documents and categories without developer involvement.

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Citizen Workflows

Registration Flow

01

Enter National ID & Personal Details

02

Phone Number Verified via OTP

03

Registration Submitted — Pending Admin Approval

04

Account Activated Upon Approval

Applying for a Document

01

Browse Catalogue by Category

02

Dynamic Form Generated from Requirements

03

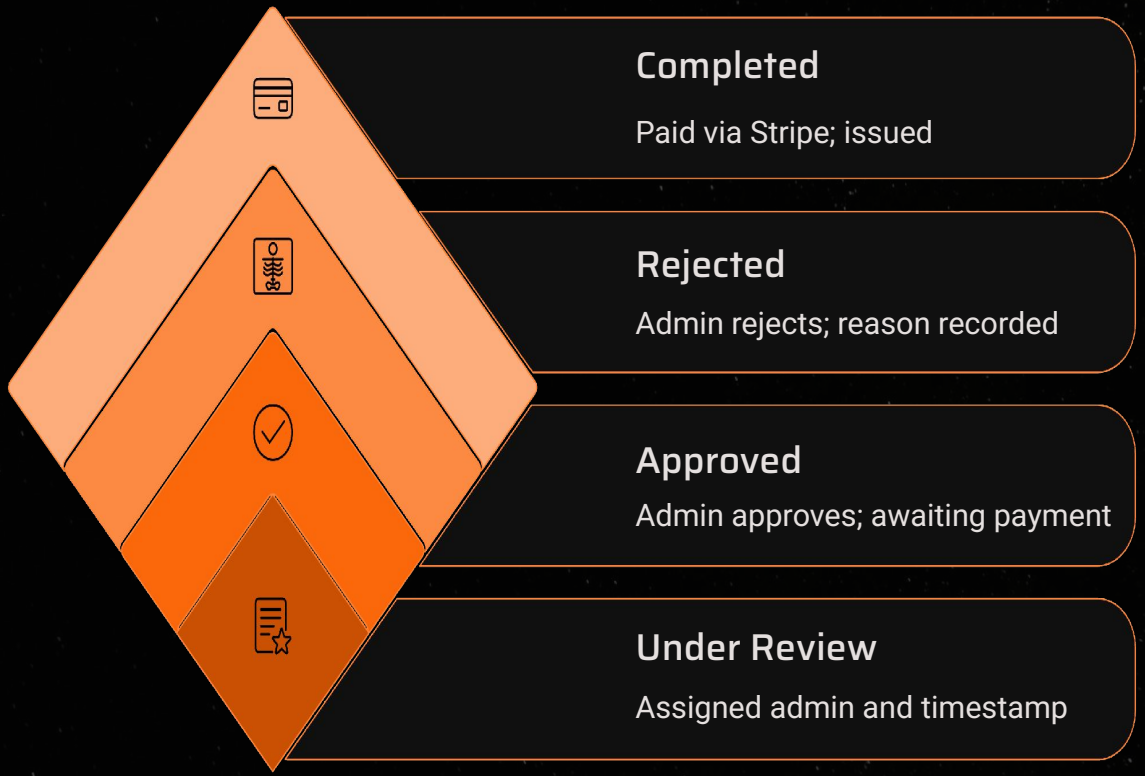
Conditional Fields Appear Based on Answers

04

Upload Files & Submit Application

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Application Lifecycle & Payment



Completed

Paid via Stripe; issued

Rejected

Admin rejects; reason recorded

Approved

Admin approves; awaiting payment

Under Review

Assigned admin and timestamp

Every administrative decision — approval or rejection — is **attributed to the specific administrator** who made it, with a recorded timestamp. Rejections carry a mandatory written reason enforced at the database level.

Payment Flow: Backend creates a Stripe `PaymentIntent` → client secret passed to mobile app → native `PaymentSheet` completes transaction → application advances to **Completed** only on confirmed payment.

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Admin Workflow & Security

Administrator Capabilities

Manage Catalogue: Create, edit, and deactivate categories, documents, and requirements — including conditional reveal rules and input types

Review Registrations: Approve or reject pending citizen registrations; rejections require a recorded reason

Review Applications: Inspect uploaded documents and form responses; approve or reject with decision attribution and timestamp

Security by Design

Argon2id Hashing

Memory-hard, OWASP-recommended — no password stored in recoverable form.

JWT Role-Based Access

Every request verified against a signed token carrying the user's role.

OTP-Gated Actions

Registration, phone changes, and password resets all require phone verification.

Enumeration-Resistant Reset

Password-reset path does not reveal whether an account exists.

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Data Integrity & Future Roadmap

Integrity rules — valid state transitions, mandatory rejection reasons, decision attribution — are enforced as **database-level constraints**, not solely in application code.



Digital Document Delivery

Cryptographically signed PDFs delivered directly through the platform.



Step-Up Authentication

Additional OTP verification for high-sensitivity documents like property registrations.



Admin Role Hierarchy

Department-level admins, system admins, and audit roles for multi-department deployment.



Push Notifications

Automatic alerts for status changes, payment due, and decisions recorded.



Analytics Dashboard

Application volumes, review times, rejection rates, and payment statistics for supervisors.



Conclusion: The platform reduces in-person attendance to the single legally necessary act of collecting the final document — a secure, data-driven foundation ready for production deployment.

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